# PerpAdapt: Radiation-free Perpendicular Pivot Calibration for Sensor-Attached C-arms with Memory Sample Consensus

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**Abstract.** Intra-operative mobile C-arm imaging provides 2D X-ray projections, whereas surgical decisions (e.g., tool trajectory, cortical breach, and implant placement) require 3D spatial reasoning. This creates substantial cognitive burden in multi-view fluoroscopy workflows. Augmented reality (AR) can reduce this burden by transferring 3D reasoning to computation, but requires accurate calibration between the attached sensor (camera/tracker/HMD) and the C-arm imaging geometry. Existing calibration methods are largely radiation-based (e.g., phantom-, CBCT-, or image-correspondence-driven), increasing workflow complexity and radiation exposure. We propose **PerpAdapt**, a radiation-free calibration framework for sensor-attached mobile C-arms. First, we show a minimal-observability result: under an orthogonal dual-axis C-arm configuration, two perpendicular rotations form a sufficient set for pivot recovery. Second, we introduce **Perpendicular Pivot Calibration (PPC)**, which directly recovers the C-arm pivot center from the common perpendicular of two fitted rotation axes, avoiding torus fitting and multi-trajectory aggregation. Third, we introduce **Memory Sample Consensus (MemorySAC)**, an ODE-guided robust estimator for stable axis fitting under noisy trajectories. On the high-fidelity benchmark, PerpAdapt achieves *order-of-magnitude* improvements in pivot accuracy, is markedly more robust under noise, and runs substantially faster than the radiation-free pivot baseline.

**Keywords:** Mobile C-arm calibration · Radiation-free calibration · Augmented reality · Neural ODE

## 1 Introduction

Mobile C-arms are widely used in orthopedic trauma, spine surgery, and instrument-guided procedures for intra-operative imaging and verification [13]. In current workflows, surgeons repeatedly inspect *2D* X-ray projections while making decisions in *3D* space, such as adjusting K-wire direction, checking implant trajectory, or verifying whether a screw breaches cortical boundaries [12]. This requires repeated mental reconstruction from multiple 2D views to an internal 3D spatial model, which is cognitively demanding and error-prone, especially for less experienced operators [4,6,5].

Augmented reality (AR) offers a principled way to reduce this burden by shifting spatial reasoning from the surgeon's mental model to the computer: instead of inferring whether a tool is correctly positioned from multiple fluoroscopic views, the system can directly present the tool and anatomy in a shared 3D reference frame [6]. To enable this, the sensing modality attached to the C-arm (e.g., RGB/RGBD camera, tracker, or head-mounted display) must be accurately calibrated to the C-arm imaging geometry, i.e., the sensor and the C-arm must be expressed in a common coordinate system [1,8].

Historically, C-arm sensor calibration has largely been *radiation-based*. Early camera-augmented mobile C-arm systems relied on image alignment and optical/X-ray correspondences [11,13]. Later methods used calibration phantoms visible in both optical and X-ray modalities [1], CBCT reconstruction followed by 3D alignment (e.g., ICP) [8], or hand-eye style formulations that still depend on knowing the imaging geometry through image/scan acquisition [15,6,14]. These approaches are effective but share a practical limitation: calibration remains tied to image-based correspondences or CBCT scans, which increases setup complexity and introduces additional radiation exposure.

Lee *et al.* [9] introduced the first radiation-free pivot calibration concept for sensor-attached mobile C-arms by exploiting constrained C-arm mechanics and tracked sensor trajectories. This is an important shift: the calibration problem is reformulated from an image-based correspondence problem into a constrained trajectory-geometry problem on $SE(3)$. However, the original formulation models the motion locus through torus geometry and estimates intermediate quantities (e.g., major radius) before recovering the pivot center, which introduces redundant estimation steps and can hurt efficiency and numerical stability.

In this paper, we propose **PerpAdapt**, a radiation-free calibration framework designed for orthogonal dual-axis mobile C-arms. Our method is built around the observation that the x-axis and C-arm-axis form an approximately perpendicular mechanical pair, which permits a more direct geometric recovery of the pivot center.

*Contributions.*

- **Minimal observability:** We show that *two perpendicular rotations form a minimal sufficient set* for radiation-free pivot recovery, reducing trajectory requirements from multi-trajectory torus aggregation to a 2-circle observable setup.
- **Perpendicular pivot calibration:** We propose a one-step pivot recovery method based on the *common perpendicular* of two fitted rotation axes. Compared with torus-based formulations, PPC uses no torus fitting, no multi-trajectory aggregation, and no iterative axis disentanglement.
- **MemorySAC:** We introduce *Memory Sample Consensus*, an ODE-guided robust estimator tailored for trajectory denoising and axis fitting stability, especially for noisy or degenerate circle samples.
- **Strong empirical performance:** On the high-fidelity benchmark [9], PerpAdapt achieves *order-of-magnitude* gains in pivot accuracy, with consis-

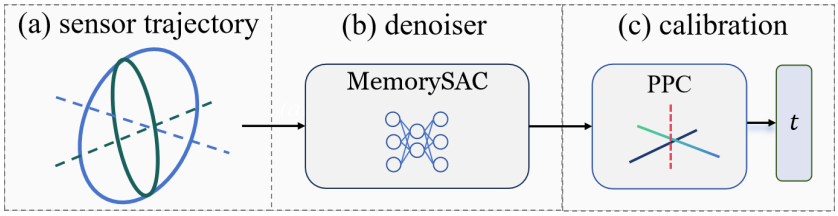



Fig. 1: Overview of PerpAdapt. Two trajectories are collected by rotating the mobile C-arm around the C-arm-axis and x-axis, respectively. MemorySAC denoises raw tracked poses, PPC fits the two rotation axes and recovers the pivot center $c_c$ from their common perpendicular, then estimates the sensor-to-center offset $t$ and downstream C-arm geometric entities.

tently stronger robustness and substantially improved runtime over the radiation-free pivot baseline.

## 2 Method

### 2.1 Problem Setup and Data Representation

We consider a sensor rigidly attached to a mobile C-arm. Let $c_c \in \mathbb{R}^3$ denote the C-arm pivot center (rotation center), and let $t \in \mathbb{R}^3$ denote the constant sensor-to-center offset expressed in the sensor frame. For each tracked sensor pose $T_i \in SE(3)$, the pivot center satisfies

$$c_c = T_i(t), \qquad t = T_i^{-1}(c_c). \tag{1}$$

*C-arm rotation axes and trajectories.* A mobile C-arm provides two principal rotations: an orbital rotation about the *x-axis* (roll around the patient) and an angulation rotation about the *C-arm-axis*. We collect two pose trajectories by actuating one axis at a time while keeping the other fixed:

- $\{\hat{T}_i^c\}_{i=1}^{n_c}$: poses acquired while rotating about the *C-arm-axis* (x-axis angle fixed);
- $\{\hat{T}_i^x\}_{i=1}^{n_x}$: poses acquired while rotating about the *x-axis* (C-arm angle fixed).

Hats indicate raw trajectories, which may contain noise and outliers from tracking or synchronization.

PerpAdapt consists of three stages (Fig. 1): (1) two-trajectory acquisition, (2) trajectory denoising via MemorySAC, and (3) PPC-based pivot recovery and C-arm geometric reconstruction.

### 2.2 Minimal Observability Under Orthogonal Dual-Axis Motion

The key geometric prior is that, in modern mobile C-arms, the x-axis and C-arm-axis are mechanically designed to be orthogonal and non-intersecting [9]. Under

this condition, each single-axis rotation yields a circle trajectory of the tracked sensor, whose fitted normal defines the corresponding rotation axis. Therefore, two rotations (one per axis) produce two axis lines $l_x$ and $l_c$, and their unique common perpendicular determines the pivot center on $l_c$.

This yields a *2-circle minimal observable set*: unlike torus-based approaches that aggregate multiple trajectories and intermediate radii, the pivot center is directly observable from two principal rotations. This is the geometric basis of PPC.

### 2.3 Perpendicular Pivot Calibration (PPC)

PPC estimates the C-arm pivot center using only two single-axis rotations (Fig. 2). It proceeds in three steps: (i) fit two rotation axes $l_x, l_c$ from the two circle trajectories, (ii) recover the pivot center $c_c$ as the foot point on $l_c$ of the common perpendicular between $l_x$ and $l_c$, and (iii) compute the sensor-to-center offset by averaging $t = \frac{1}{n_c} \sum_i (T_i^c)^{-1}(c_c)$.

*Axis fitting.* From denoised trajectories, we extract 3D point samples and fit two circles, one from the x-axis rotation and one from the C-arm-axis rotation. The fitted circle normals define the axis directions, producing two lines:

$$l_x : \mathbf{r}_x = \mathbf{a}_x + u\mathbf{d}_x, \qquad l_c : \mathbf{r}_c = \mathbf{a}_c + v\mathbf{d}_c, \tag{2}$$

where $\mathbf{a}_x, \mathbf{a}_c$ are points on the lines and $\mathbf{d}_x, \mathbf{d}_c$ are direction vectors.

*One-step pivot recovery from the common perpendicular.* For two skew lines, the shortest connecting segment is perpendicular to both. Thus $(\mathbf{r}_x - \mathbf{r}_c)$ is orthogonal to $\mathbf{d}_x$ and $\mathbf{d}_c$, yielding:

$$(\mathbf{a}_x - \mathbf{a}_c + u\mathbf{d}_x - v\mathbf{d}_c) \cdot \mathbf{d}_x = 0, \tag{3}$$

$$(\mathbf{a}_x - \mathbf{a}_c + u\mathbf{d}_x - v\mathbf{d}_c) \cdot \mathbf{d}_c = 0. \tag{4}$$

This gives the $2 \times 2$ linear system

$$\begin{bmatrix} \mathbf{d}_x \cdot \mathbf{d}_x & -\mathbf{d}_x \cdot \mathbf{d}_c \\ \mathbf{d}_x \cdot \mathbf{d}_c & -\mathbf{d}_c \cdot \mathbf{d}_c \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} -(\mathbf{a}_x - \mathbf{a}_c) \cdot \mathbf{d}_x \\ -(\mathbf{a}_x - \mathbf{a}_c) \cdot \mathbf{d}_c \end{bmatrix}. \tag{5}$$

After solving Eq. (5), the pivot center is directly recovered as the foot point on the C-arm-axis:

$$c_c = \mathbf{a}_c + v\mathbf{d}_c. \tag{6}$$

Compared with torus-based pivot calibration [9], PPC eliminates major-radius estimation and multi-step geometric disentanglement. This reduces process redundancy and improves numerical stability in practice.

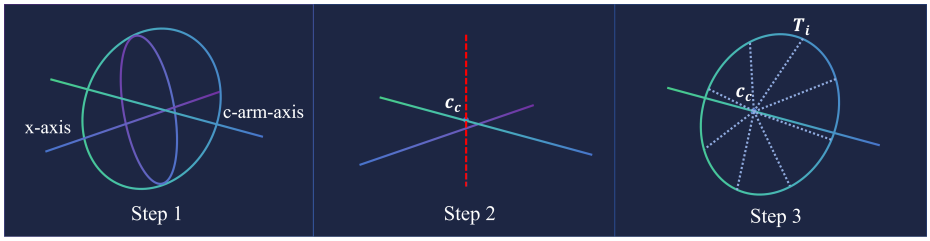



Fig. 2: Perpendicular Pivot Calibration (PPC). Step 1: two single-axis rotation trajectories are used to fit the axis lines $l_x$ and $l_c$. Step 2: the pivot center $c_c$ is recovered from the relative geometry of the two axes (common-perpendicular foot point). Step 3: $t$ is computed by aggregating $(T_i^c)^{-1}(c_c)$ over poses.

*Sensor-to-center offset estimation.* Once $c_c$ is known, the sensor-to-center offset is obtained by averaging inverse-transformed pivot locations:

$$t = \frac{1}{n_c} \sum_{i=1}^{n_c} (T_i^c)^{-1}(c_c). \tag{7}$$

This is the core calibration output for sensor-to-C-arm registration.

*Recovering C-arm emission direction and imaging plane.* Beyond $t$, downstream AR usage requires a consistent C-arm geometric direction to infer the X-ray emission axis and detector plane. After obtaining $c_c$, we compute its orthogonal projection onto the fitted x-axis line $l_x$, denoted by $q$. The vector from $q$ to $c_c$ is perpendicular to the x-axis and defines the C-arm local $z$-axis:

$$\mathbf{z} = \frac{c_c - q}{\|c_c - q\|}, \tag{8}$$

with the sign chosen to point *away from the x-axis* (device-consistent convention). The $y$-axis is not required in our calibration output. Given the calibrated device geometry (source-detector offsets in the C-arm local frame), the X-ray source point and detector plane can then be recovered by rigidly placing them along $\mathbf{z}$ through the pivot-centered C-arm frame. This makes the PPC output directly usable for AR overlay and tool visualization.

### 2.4 Memory Sample Consensus (MemorySAC)

**Why not plain RANSAC.** PPC recovers the pivot center from the common perpendicular of two fitted axes, hence axis-fitting errors can be amplified in the pivot estimate. With tracked C-arm trajectories, outliers are often *structured* (e.g., clustered glitches, short/uneven arcs), so the inlier probability varies along the circle. Under a small iteration budget, uniform RANSAC wastes hypotheses on low-quality regions and can yield unstable circle normals. We therefore use a *score-guided* consensus procedure to reliably obtain stable circle fits for PPC.

*Core idea.* MemorySAC keeps the standard RANSAC scoring rule (inlier thresholding) but replaces *uniform sampling* with *weighted sampling*. Each point $\mathbf{x}_i$ is assigned a reliability score $P_i \in (0, 1)$; points with larger $P_i$ are sampled more frequently, increasing the probability of drawing an outlier-free minimal set early.

*Reliability scores via ODE.* We learn $P_i$ using an ODE-based memory scorer [2,16,10], which provides stable, noise-tolerant representations through dissipative continuous dynamics. The scorer maps a point embedding $\phi(\mathbf{x}_i)$ to a probability $P_i$ (details in implementation).

*Algorithm (per trajectory).* Given 3D samples $\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^{N}$, MemorySAC returns an inlier set $\mathcal{I}$ used for circle/axis fitting:

1. **Score:** compute $P_i = \text{MemoryScorer}(\mathbf{x}_i)$ for all points.
2. **Sample:** repeat for $K$ iterations: sample a minimal set (3 points) with $\Pr(i) \propto P_i$, fit a circle hypothesis $\mathcal{C}$.
3. **Consensus:** compute residuals $r_i = \text{dist}(\mathbf{x}_i, \mathcal{C})$, form inliers $\mathcal{I}(\mathcal{C}) = \{i \mid r_i < \tau\}$.
4. **Select:** keep the hypothesis with the largest $|\mathcal{I}(\mathcal{C})|$ (tie-break by smaller $\sum_{i\in\mathcal{I}} r_i$); output $\mathcal{I} = \mathcal{I}(\mathcal{C}^*)$.

MemorySAC is applied independently to the x-trajectory and the C-arm-axis trajectory, and PPC fits $l_x, l_c$ from the resulting inliers. In practice, this improves pivot recovery by stabilizing circle normals under structured outliers.

## 3 Experiments

### 3.1 Experimental Setup and Evaluation Protocol

We follow the benchmark and evaluation protocol of Lee *et al.* [9], using realistic mobile C-arm geometry/noise ranges informed by clinical-device constraints. We generate 10,000 calibration instances, each containing two trajectories with $K = 100$ poses per circle by default, and use 5-fold cross-validation for learned components. We evaluate the estimated offset $t$ (Eq. 7) by MAE/MSE/$R^2$ averaged over its 3D components. Unless stated otherwise, we compare against the radiation-free pivot baseline [9] in both its 4-circle (Lee (4)) and 6-circle (Lee (6)) variants under three settings:

- **Accuracy:** varying points per circle $(10, 20, 50, 100)$;
- **Robustness:** Gaussian/uniform perturbations on randomly selected trajectory points;
- **Runtime:** average time per calibration instance.

All methods are implemented in PyTorch 1.13.1 on an RTX 3090; MemoryScorer is trained for 100 epochs using Adam [7] (lr $10^{-3}$, batch 256). RANSAC/MemorySAC uses 10 iterations, sample size 3, and threshold 0.05.

Table 1: Accuracy comparison across point counts. Best values are bold.

| Method | Point # | Metrics | | |
|---|---|---|---|---|
| | | MAE | MSE | $R^2$ |
| Lee (4) [9] | 10 | $3.88 \times 10^{-3} \pm 1.97 \times 10^{-3}$ | $6.64 \times 10^{-5} \pm 5.81 \times 10^{-5}$ | $0.999859 \pm 1.30 \times 10^{-4}$ |
| | 20 | $4.08 \times 10^{-3} \pm 1.83 \times 10^{-3}$ | $7.09 \times 10^{-5} \pm 5.85 \times 10^{-5}$ | $0.999827 \pm 1.76 \times 10^{-4}$ |
| | 50 | $4.20 \times 10^{-3} \pm 2.29 \times 10^{-3}$ | $8.07 \times 10^{-5} \pm 7.81 \times 10^{-5}$ | $0.999812 \pm 1.88 \times 10^{-4}$ |
| | 100 | $4.13 \times 10^{-3} \pm 2.10 \times 10^{-3}$ | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $0.999813 \pm 2.02 \times 10^{-4}$ |
| Lee (6) [9] | 10 | $3.88 \times 10^{-3} \pm 1.97 \times 10^{-3}$ | $6.64 \times 10^{-5} \pm 5.81 \times 10^{-5}$ | $0.999859 \pm 1.30 \times 10^{-4}$ |
| | 20 | $4.08 \times 10^{-3} \pm 1.83 \times 10^{-3}$ | $7.09 \times 10^{-5} \pm 5.85 \times 10^{-5}$ | $0.999841 \pm 1.88 \times 10^{-4}$ |
| | 50 | $4.20 \times 10^{-3} \pm 2.29 \times 10^{-3}$ | $8.07 \times 10^{-5} \pm 7.81 \times 10^{-5}$ | $0.999812 \pm 1.88 \times 10^{-4}$ |
| | 100 | $4.13 \times 10^{-3} \pm 2.10 \times 10^{-3}$ | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $0.999813 \pm 2.02 \times 10^{-4}$ |
| **PerpAdapt (Ours)** | **10** | $\mathbf{4.76 \times 10^{-4} \pm 6.39 \times 10^{-5}}$ | $\mathbf{3.97 \times 10^{-7} \pm 9.36 \times 10^{-8}}$ | $\mathbf{0.999999 \pm 1.94 \times 10^{-7}}$ |
| | **20** | $\mathbf{4.73 \times 10^{-4} \pm 5.69 \times 10^{-5}}$ | $\mathbf{3.87 \times 10^{-7} \pm 9.10 \times 10^{-8}}$ | $\mathbf{0.999999 \pm 1.85 \times 10^{-7}}$ |
| | **50** | $\mathbf{4.71 \times 10^{-4} \pm 5.21 \times 10^{-5}}$ | $\mathbf{3.85 \times 10^{-7} \pm 9.06 \times 10^{-8}}$ | $\mathbf{0.999999 \pm 1.85 \times 10^{-7}}$ |
| | **100** | $\mathbf{4.70 \times 10^{-4} \pm 5.03 \times 10^{-5}}$ | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ | $\mathbf{0.999999 \pm 1.86 \times 10^{-7}}$ |

Table 2: Robustness comparison (MSE) under representative noise settings. Best values are bold.

| Noise Type | Direction | Noise Level | Lee (4) | Lee (6) | **PerpAdapt** |
|---|---|---|---|---|---|
| Gaussian | $l_c$ | $\delta \sim \mathcal{N}(0,1)$ | $6.85 \times 10^{-5} \pm 6.66 \times 10^{-5}$ | $6.83 \times 10^{-5} \pm 6.68 \times 10^{-5}$ | $\mathbf{6.35 \times 10^{-6} \pm 3.86 \times 10^{-6}}$ |
| Gaussian | $l_c$ | $\delta \sim \mathcal{N}(0,3)$ | $6.83 \times 10^{-5} \pm 6.68 \times 10^{-5}$ | $6.83 \times 10^{-5} \pm 6.68 \times 10^{-5}$ | $\mathbf{4.74 \times 10^{-6} \pm 1.89 \times 10^{-6}}$ |
| Gaussian | $l_x$ | $\delta \sim \mathcal{N}(0,1)$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ |
| Gaussian | $l_x$ | $\delta \sim \mathcal{N}(0,3)$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ |
| Uniform | $l_c$ | $\delta \sim \mathcal{U}(-1,1)$ | $6.84 \times 10^{-5} \pm 6.66 \times 10^{-5}$ | $6.83 \times 10^{-5} \pm 6.68 \times 10^{-5}$ | $\mathbf{4.74 \times 10^{-6} \pm 1.89 \times 10^{-6}}$ |
| Uniform | $l_c$ | $\delta \sim \mathcal{U}(-3,3)$ | $6.83 \times 10^{-5} \pm 6.68 \times 10^{-5}$ | $6.84 \times 10^{-5} \pm 6.70 \times 10^{-5}$ | $\mathbf{4.74 \times 10^{-6} \pm 1.89 \times 10^{-6}}$ |
| Uniform | $l_x$ | $\delta \sim \mathcal{U}(-1,1)$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ |
| Uniform | $l_x$ | $\delta \sim \mathcal{U}(-3,3)$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $7.73 \times 10^{-5} \pm 7.07 \times 10^{-5}$ | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ |

### 3.2 Main Results

*Accuracy.* Table 1 shows that PerpAdapt consistently outperforms both baseline variants across point counts. The gains are substantial: MAE improves by approximately one order of magnitude, and MSE by approximately two orders of magnitude. This validates the central premise of PPC: directly estimating $c_c$ from the common perpendicular is more stable than recovering it through torus-based intermediate geometry.

*Robustness under noise.* Table 2 summarizes robustness to Gaussian and uniform perturbations. PerpAdapt consistently yields the lowest MSE, with the largest gains under perturbations that affect axis fitting. This supports the role of MemorySAC in improving the stability of circle fitting and common-perpendicular estimation.

*Runtime.* Table 3 shows that PerpAdapt is substantially faster than both baseline variants. The speedup follows directly from the minimal-observable formulation (two trajectories only) and the one-step PPC recovery, while MemorySAC reduces ineffective sampling in noisy conditions.

Table 3: Runtime comparison (seconds). Best values are bold.

| Method | Point Number | | | |
|---|---|---|---|---|
| | 10 | 20 | 50 | 100 |
| Lee (4) [9] | $0.447 \pm 0.004$ | $0.451 \pm 0.004$ | $0.603 \pm 0.003$ | $0.858 \pm 0.005$ |
| Lee (6) [9] | $0.484 \pm 0.006$ | $0.510 \pm 0.003$ | $0.696 \pm 0.008$ | $1.017 \pm 0.013$ |
| **PerpAdapt (Ours)** | $\mathbf{0.060 \pm 0.005}$ | $\mathbf{0.077 \pm 0.003}$ | $\mathbf{0.121 \pm 0.002}$ | $\mathbf{0.197 \pm 0.005}$ |

Table 4: Ablation Study Results. The best results in each group are highlighted in **bold**, and the overall best results are shaded in gray.

| Method | MSE | | Speed (s) |
|---|---|---|---|
| | clean | noise | |
| **Calibration only** | | | |
| Lee (4) [9] w/o RANSAC [3] | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $9.88 \times 10^{-4} \pm 1.12 \times 10^{-4}$ | $1.067 \pm 0.024$ |
| Lee (6) [9] w/o RANSAC [3] | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $\mathbf{1.68 \times 10^{-3} \pm 6.95 \times 10^{-5}}$ | $1.326 \pm 0.029$ |
| **PerpAdapt** w/o MemorySAC | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ | $5.68 \times 10^{-3} \pm 2.33 \times 10^{-5}$ | $\mathbf{0.135 \pm 0.004}$ |
| **Calibration+RANSAC** | | | |
| Lee (4) [9] | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $2.30 \times 10^{-3} \pm 4.81 \times 10^{-4}$ | $0.921 \pm 0.012$ |
| Lee (6) [9] | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $\mathbf{2.59 \times 10^{-4} \pm 1.79 \times 10^{-4}}$ | $1.095 \pm 0.015$ |
| **PerpAdapt** w/ RANSAC [3] | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ | $5.44 \times 10^{-4} \pm 2.72 \times 10^{-4}$ | $\mathbf{0.173 \pm 0.004}$ |
| **Calibration+MemorySAC** | | | |
| Lee (4) [9] w/ MemorySAC | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $9.41 \times 10^{-5} \pm 7.31 \times 10^{-5}$ | $0.996 \pm 0.008$ |
| Lee (6) [9] w/ MemorySAC | $7.75 \times 10^{-5} \pm 7.10 \times 10^{-5}$ | $9.31 \times 10^{-5} \pm 7.22 \times 10^{-5}$ | $1.197 \pm 0.012$ |
| **PerpAdapt (full)** | $\mathbf{3.84 \times 10^{-7} \pm 9.04 \times 10^{-8}}$ | $\mathbf{2.65 \times 10^{-5} \pm 1.61 \times 10^{-5}}$ | $\mathbf{0.157 \pm 0.005}$ |

### 3.3 Ablation Study

We ablate the contributions of PPC and MemorySAC using three groups: (i) calibration only, (ii) calibration + standard RANSAC, and (iii) calibration + MemorySAC. For noisy ablation, we inject Gaussian noise $\delta \sim \mathcal{N}(0, 1)$ into 50 randomly selected points per circle.

Table 4 shows that PPC alone already yields strong gains on clean data, confirming the benefit of the perpendicular pivot formulation. Under noise, standard RANSAC improves stability but still underperforms MemorySAC. The full PerpAdapt model (PPC + MemorySAC) achieves the best noisy MSE with the lowest runtime among robust variants, indicating that geometric simplification and memory-guided consensus are complementary.

## 4 Conclusion

We present **PerpAdapt**, a radiation-free calibration framework for sensor-attached mobile C-arms. **PPC** recovers the pivot center from the common perpendicular of two fitted axes without torus-based intermediate estimation, and **MemorySAC** stabilizes axis fitting under noisy trajectories. PerpAdapt consistently improves pivot accuracy, robustness, and runtime over the prior baselines. Future work will focus on integration into AR-assisted navigation systems.

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